
BY 533: Advanced Environmental Biotechnology

Course Code: BY 533

Course Name: Advanced Environmental Biotechnology

L-T-P-C: 3-0-0-3

Prerequisites: Understanding Biotechnology and its Applications (IC136) or equivalent or Permission from instructor

Students intended for: B.Tech. Bioengineering, MS/MSc, M.Tech, and PhD

Elective or Core:

Mutual Exclusion: -

Approval: 59th BoA

Course Contents

- **Module I:** Introduction to Environmental Biotechnology; Microbial Technology, Water and Wastewater Treatment Technologies: Environmental Biotechnology in Wastewater Treatment Resource Recovery from Wastewater. [6 Lectures]
- **Module II:** Sludge treatment; Bioreactors for Wastewater Treatment; Microbial and Decentralized Wastewater Treatment Technologies. [6 Lectures]
- **Module III:** Overview of Solid Waste; Environmental Biotechnology in Solid Waste Resource Recovery from Solid Waste and Sustainable Agricultural Practices, Biochar for Environmental Management and Biogas Production, Environmental Biotechnology in Plastic Waste Management; Biotechnology Application in Bioplastics, Biofuel; Biofertilizers and Biodiversity conservation. [10 Lectures]
- **Module IV:** Bioremediation and Phytoremediation, Biodegradation of Natural Compounds and Lignin biodegradation, Advancement in Hazardous Wastes Management; Biodegradation of Xenobiotic Compounds and Nanotechnology in bioremediation, Oil Spill Cleanup; Mining and Bioleaching; Value-Added Products from Organic Wastes; Biosensors and Microbes in GHG mitigation. [10 Lectures]
- **Module V:** Applications of Nanotechnology and Latest advanced technologies for environmental remediation of Wastewater, solid waste etc., Sustainable Environmental Biotechnology, Current practices in Wastewater Treatment: examples and case studies. [10 Lectures]

Textbooks:

1. McCarty, Perry L., Rittmann, Bruce E., **Environmental Biotechnology: principles and applications**, McGraw Hill, 2015.
2. Singh, A., Ward, O.P., **Applied Bioremediation and Phytoremediation**, Springer, 2004.

References:

1. IGNOU, **Environmental Biotechnology** (MEVE 013), IGNOU, ISBN 978-93-6106-062-5, 2024
2. Ulrich, A, Reuter, S., Gutterer, B., **Decentralised Wastewater Treatment Systems (DEWATS) and Sanitation in Developing Countries**, Bremen Overseas Research and Development Association (BORDA), 2009.



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3. ElMekawy, A., Mohanakrishna, G., Srikanth, S., Pant, D., [Edited by Chandra, R.], **The Role of Bioreactors in Industrial Wastewater Treatment in Environmental Waste Management**, CRC Press, 2016.